

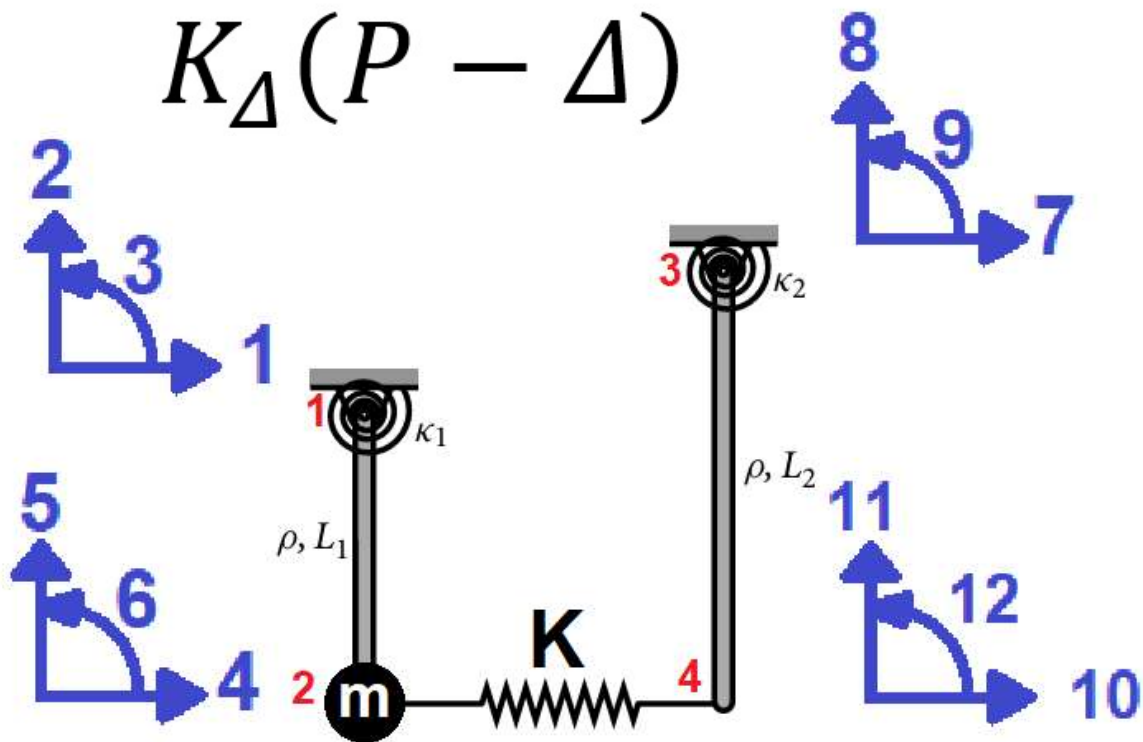
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# MODELING OF PENDULUM MDOF STRUCTURE USING OPENSEES

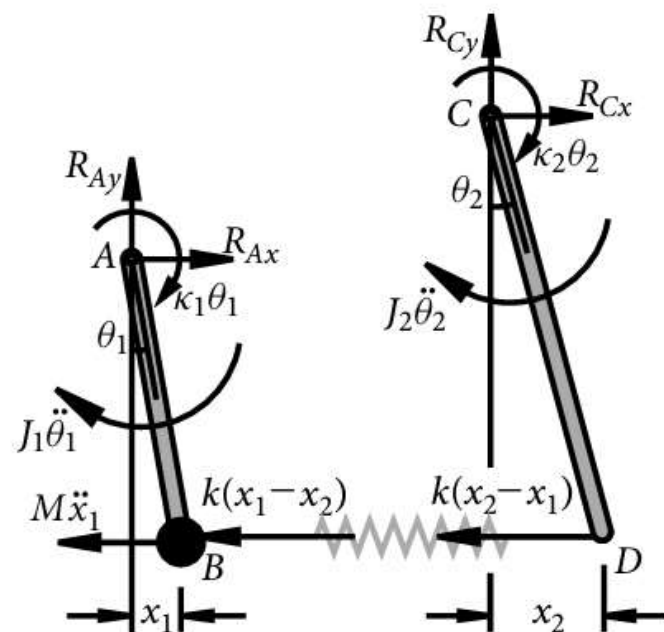
WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

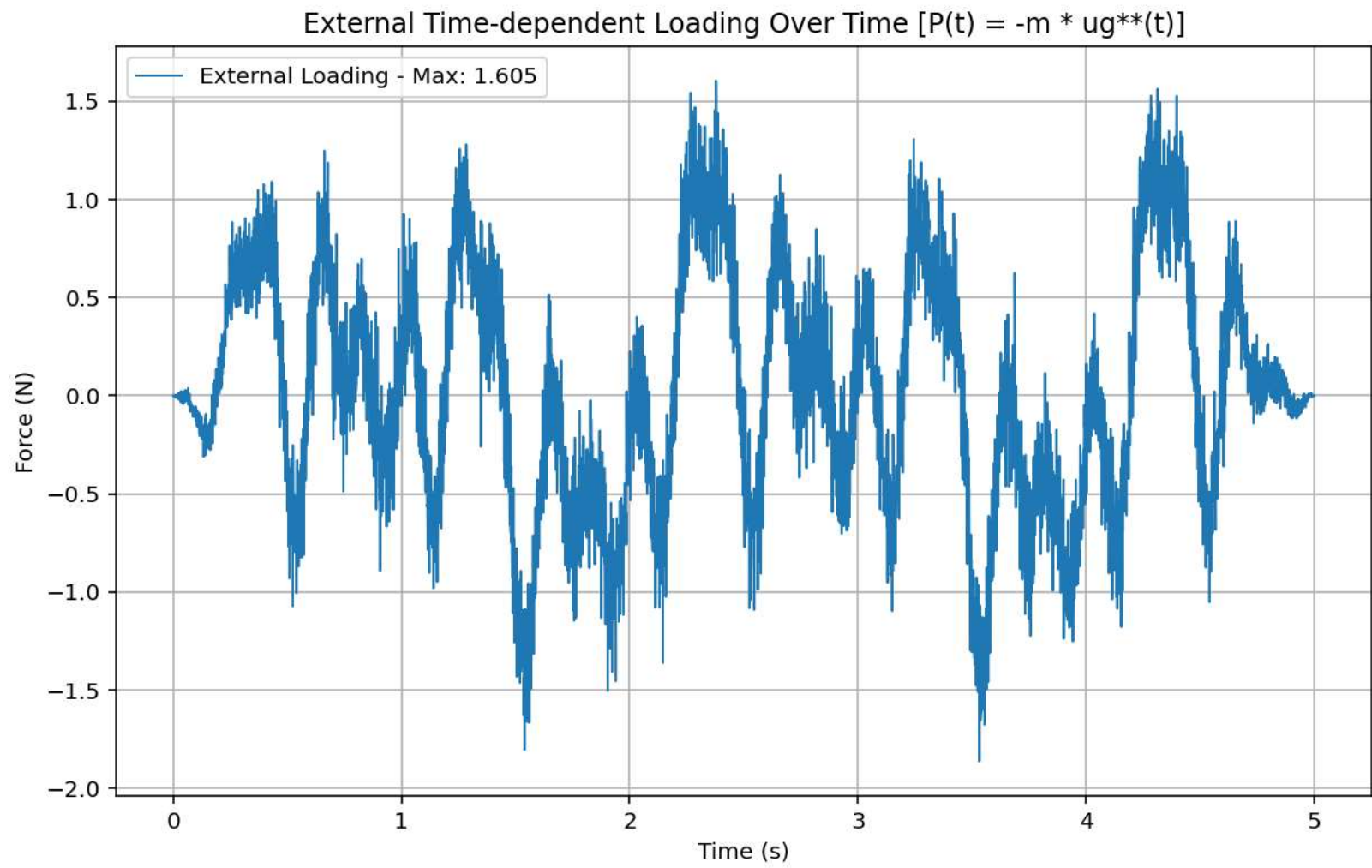
$$K_{\theta}(M - \theta)$$

$$K_{\Delta}(P - \Delta)$$



$$P(t) = -m\ddot{u}_g(t)$$





12345678910111213141516171819202122232425262728293031323334

```
#####
#                                     >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#                                     MODELING OF PENDULUM MDOF STRUCTURE USING OPENSEES
#                                     P(t) = -m * ug**(t)
#-----
#                                     EVALUATION OF DAMPING FORCE (fD), SPRING FORCE (fS) AND INERTIA FORCE (
#-----
#                                     THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                     EMAIL: salar.d.ghashghaei@gmail.com
#####
Performs time-dependent loading analysis of a Multi Degree of Freedom (MDOF)
structure using OpenSeesPy, comparing elastic and inelastic spring behavior.
Key features include:

1. Implements both elastic (linear) and hysteretic (nonlinear) material models for
structural springs.
2. Supports initial conditions for displacement, velocity, and acceleration.
3. Uses Newmark's method for time integration with Newton-Raphson iteration.
4. Calculates damping ratios using logarithmic decrement from response peaks.
5. Generates force-displacement backbone curves for inelastic material.
6. Tracks and plots time-history responses (displacement, velocity, acceleration, reactions)
7. Compares elastic vs inelastic system performance.
8. Includes convergence checks and analysis stability monitoring.
9. Outputs model data in JSON format for post-processing.
10. Provides theoretical validation through natural frequency calculations.

Particularly useful for earthquake engineering applications,
allowing evaluation of structural response under time-dependent loading
with different material nonlinearities and damping characteristics.
The hysteretic material model captures energy dissipation
inelastic deformation, while the elastic case serves as a reference for linear behavior.
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This code performs a comparative nonlinear dynamic analysis of a multi-degree-of-freedom (M
```

C:\Users\ DELL\Desktop\OPENSEES\_FILES\PENDULUM\+EXAMPLE\_05

MDOF\_PENDULUM\_FIVE.py

17 %

IPython Console

Files

Help

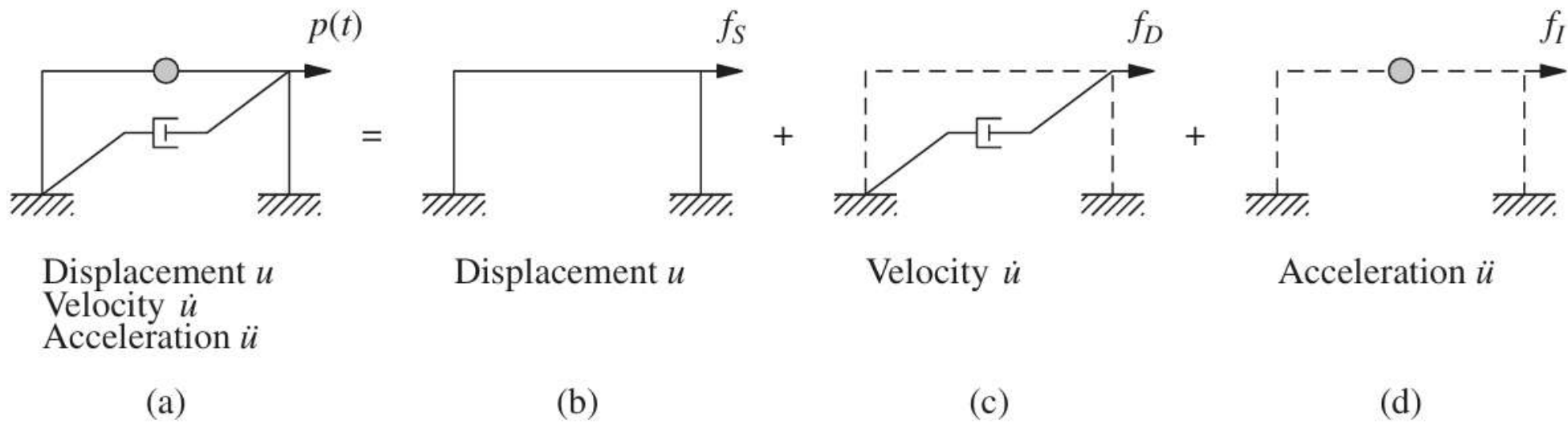
Variable Explorer

Debugger

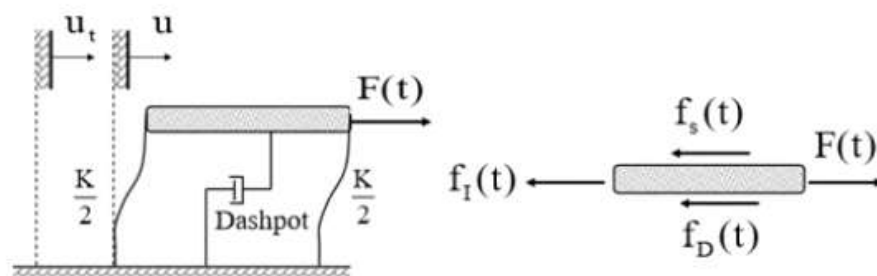
Plots

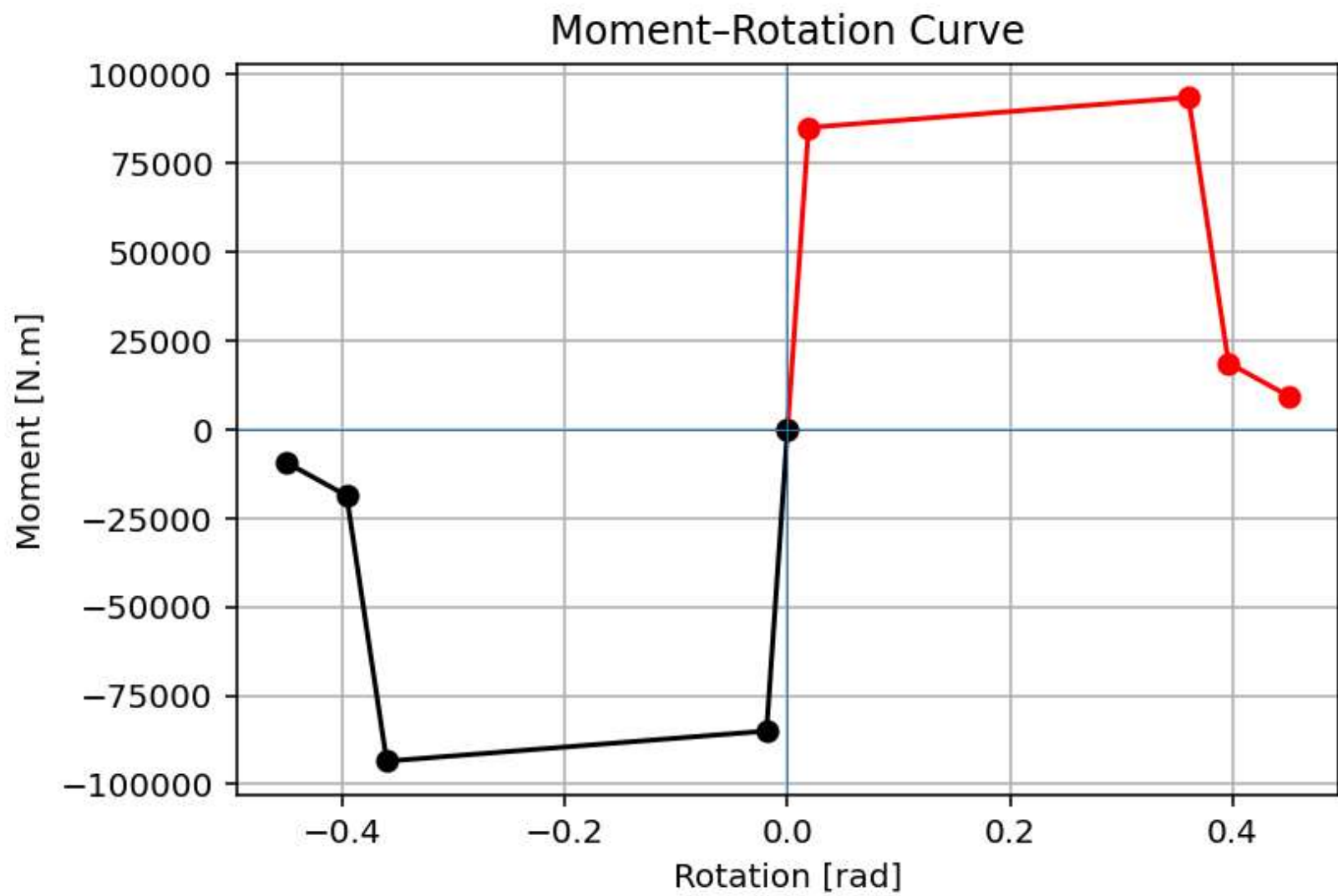
History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 311, Col 33 UTF-8 CRLF RW Mem 41%

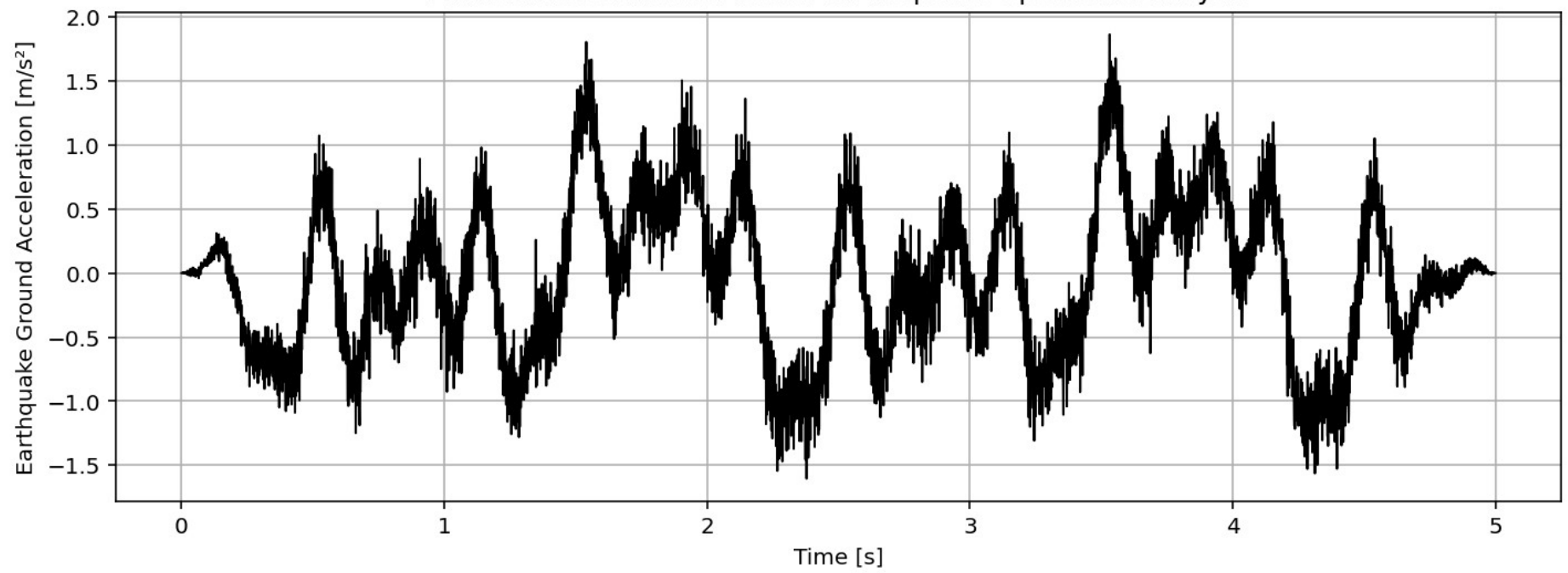


(a) System; (b) stiffness component; (c) damping component; (d) mass component.

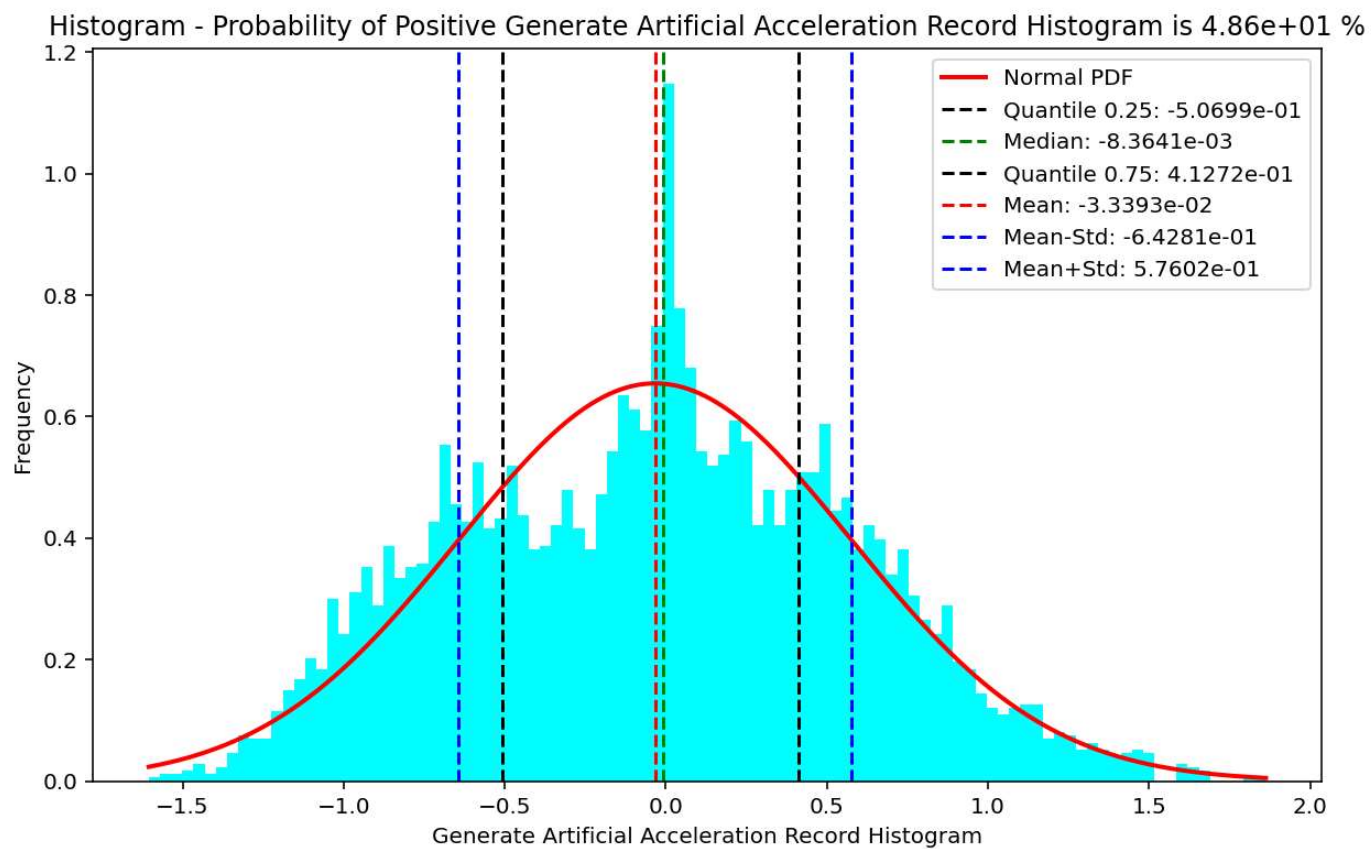




Artificial Acceleration Record for Response Spectrum Analysis

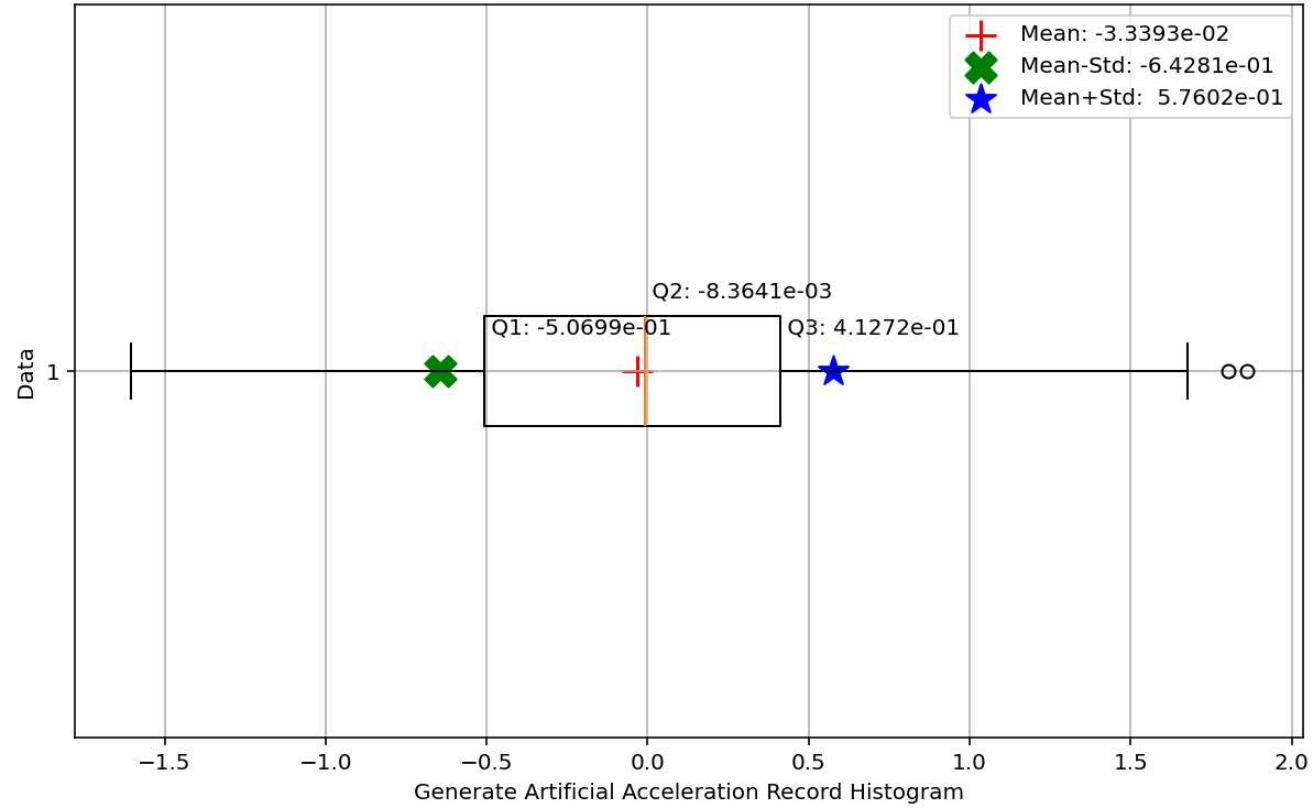




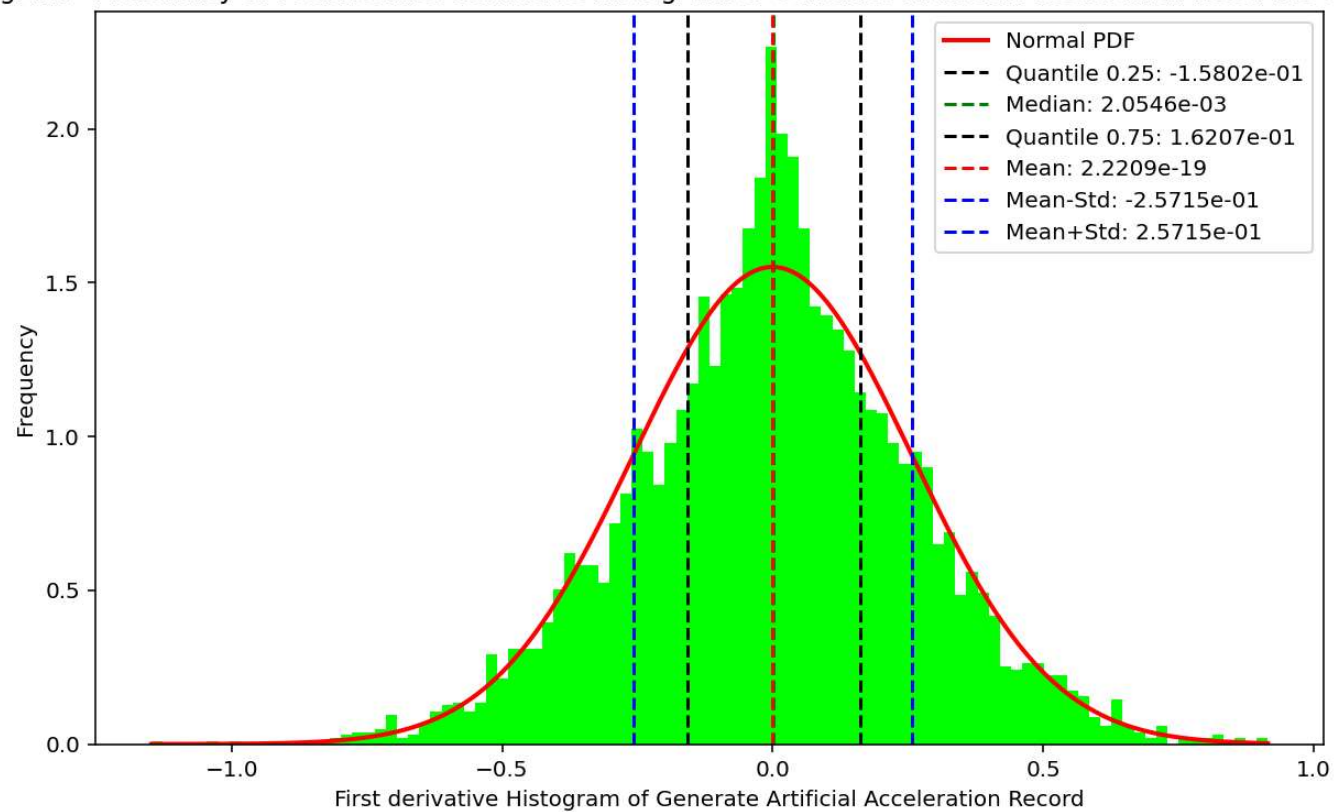


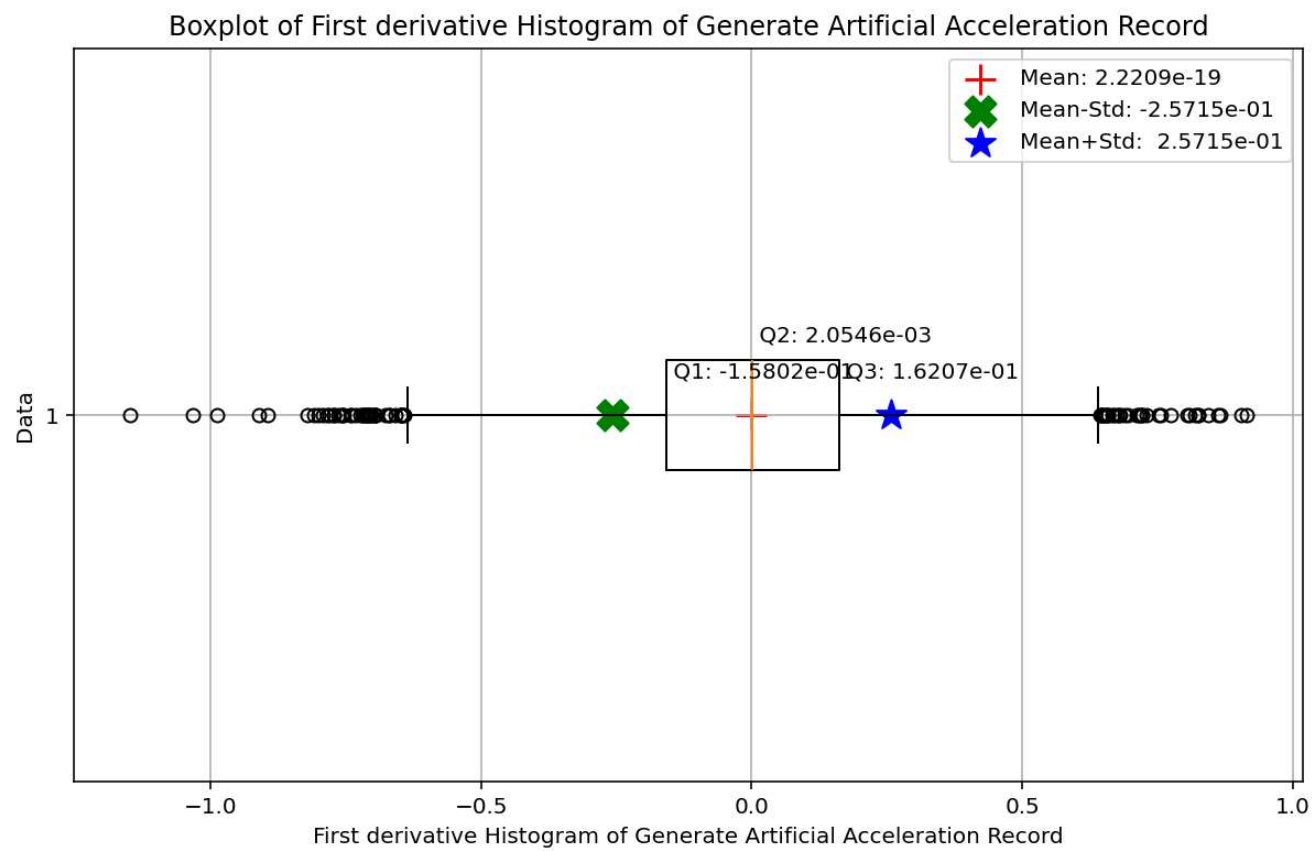


Boxplot of Generate Artificial Acceleration Record Histogram

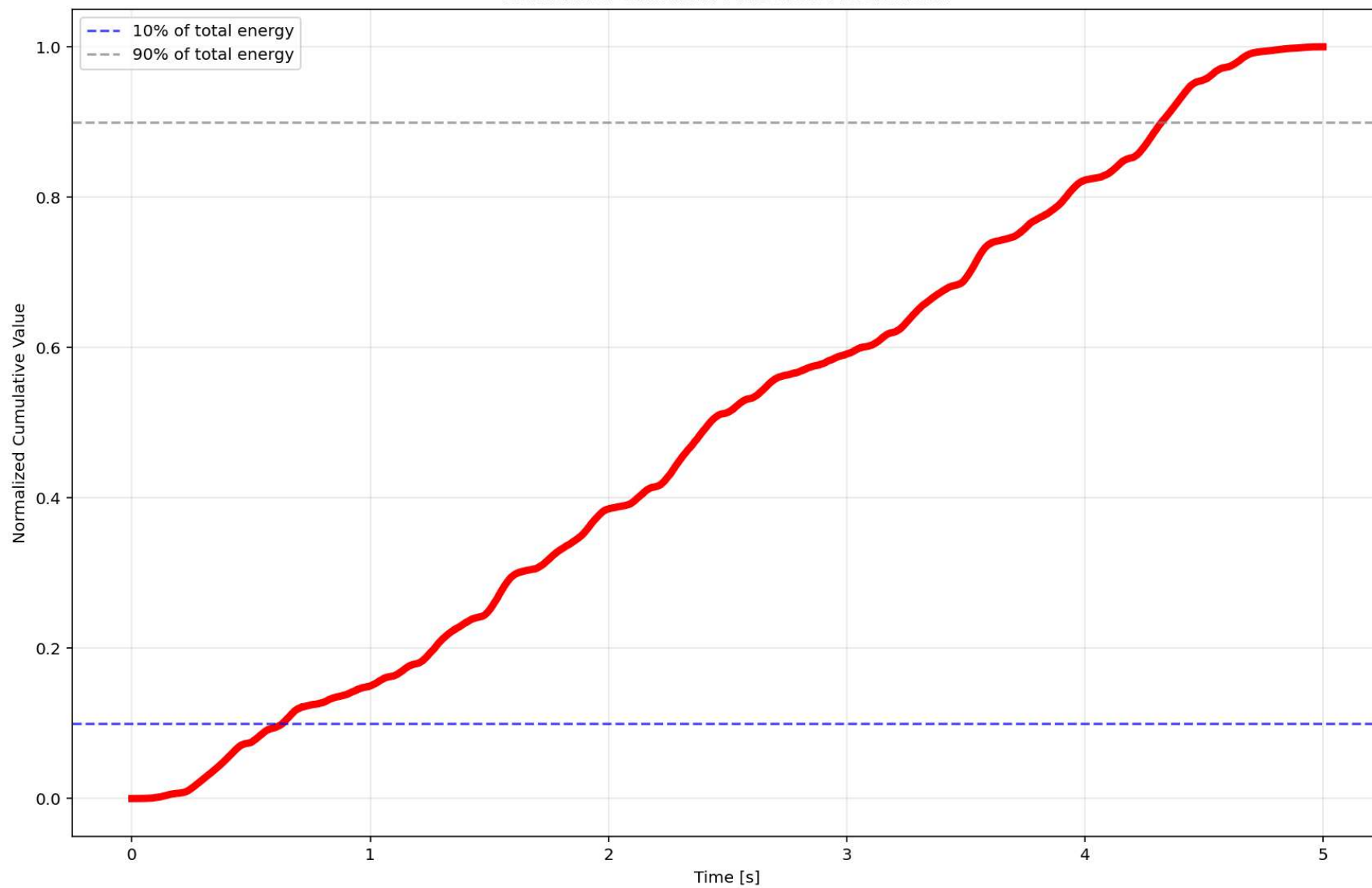


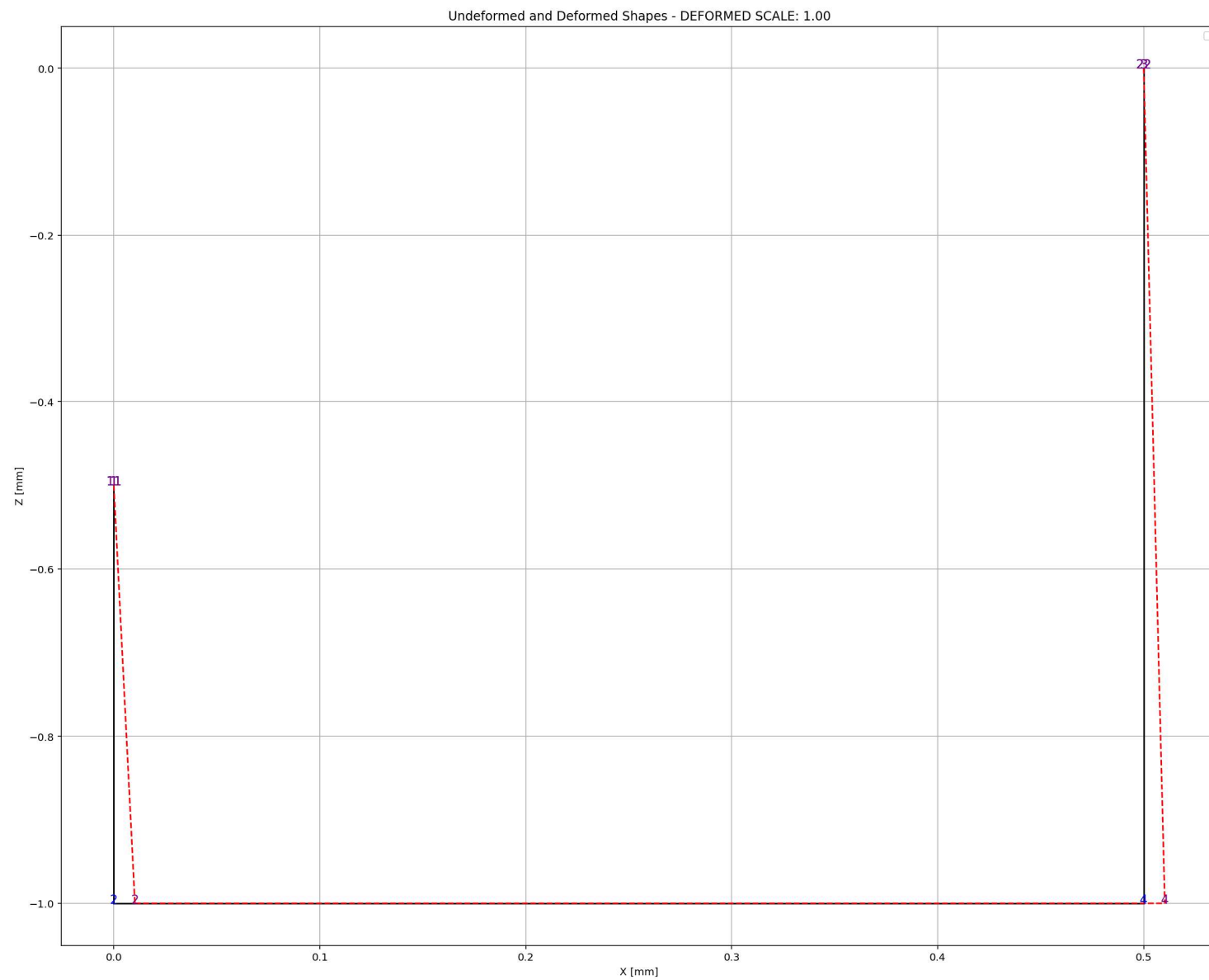
Histogram - Probability of Positive First derivative Histogram of Generate Artificial Acceleration Record is 5.07e+01 %

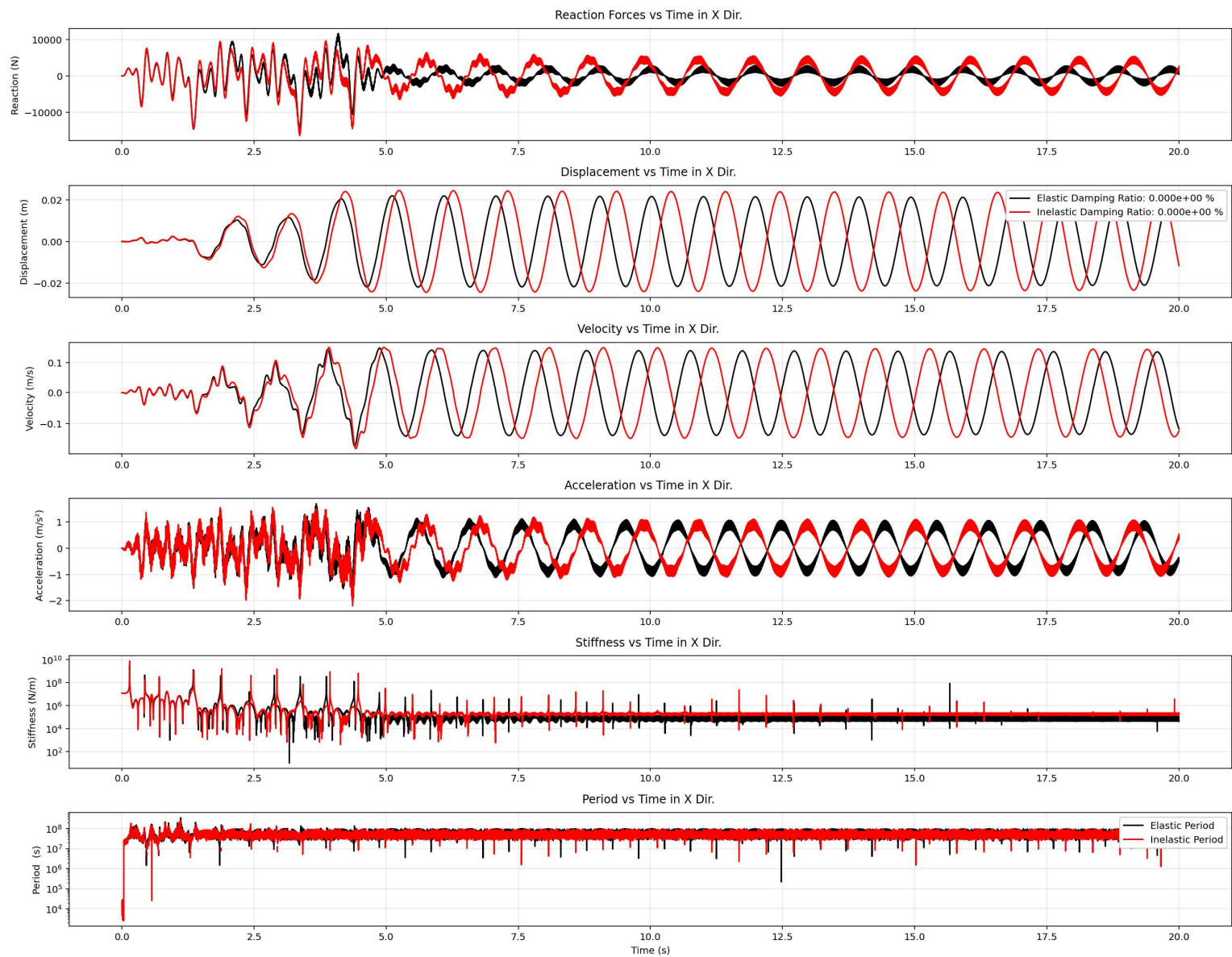


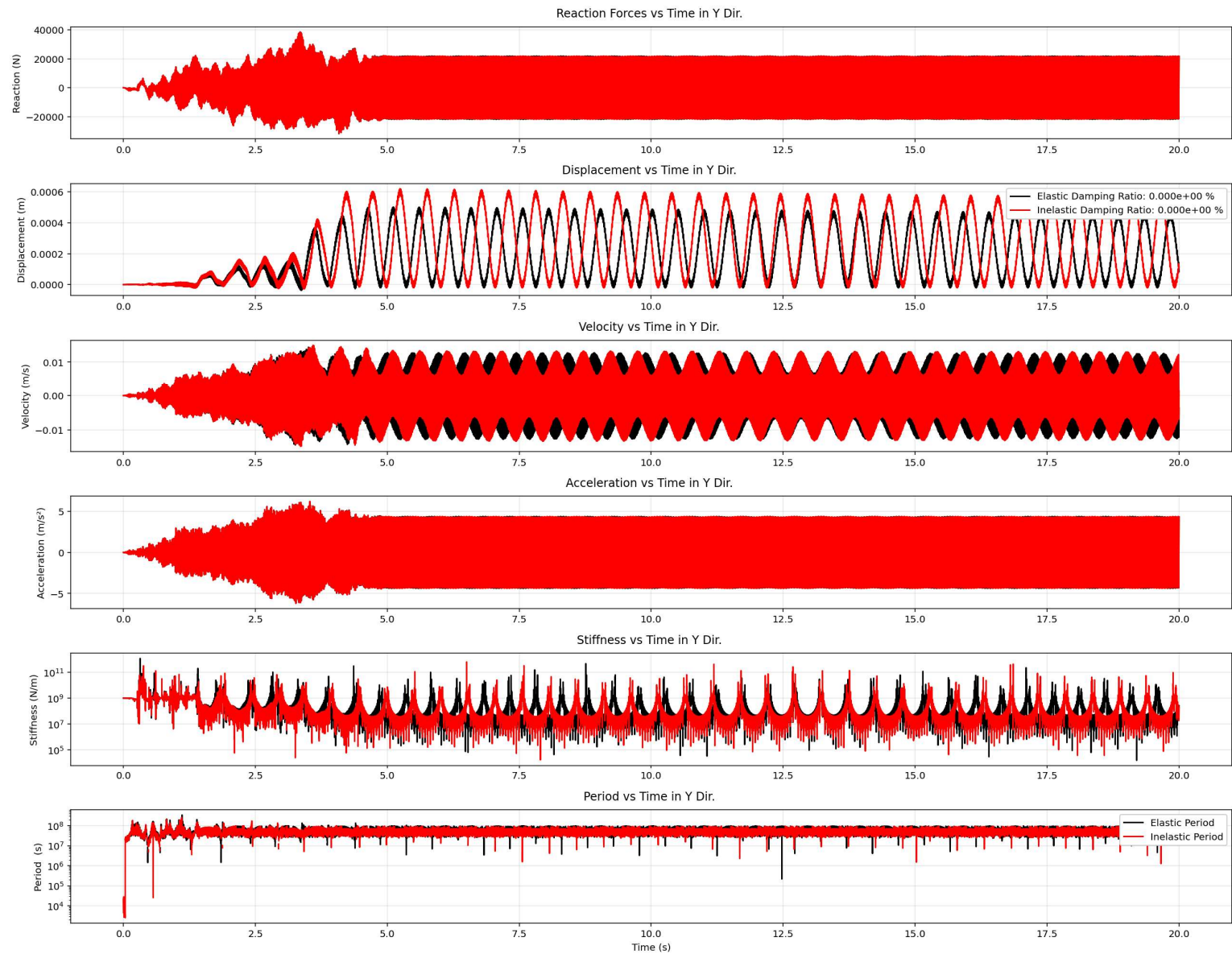


Normalized Cumulative Absolute Acceleration

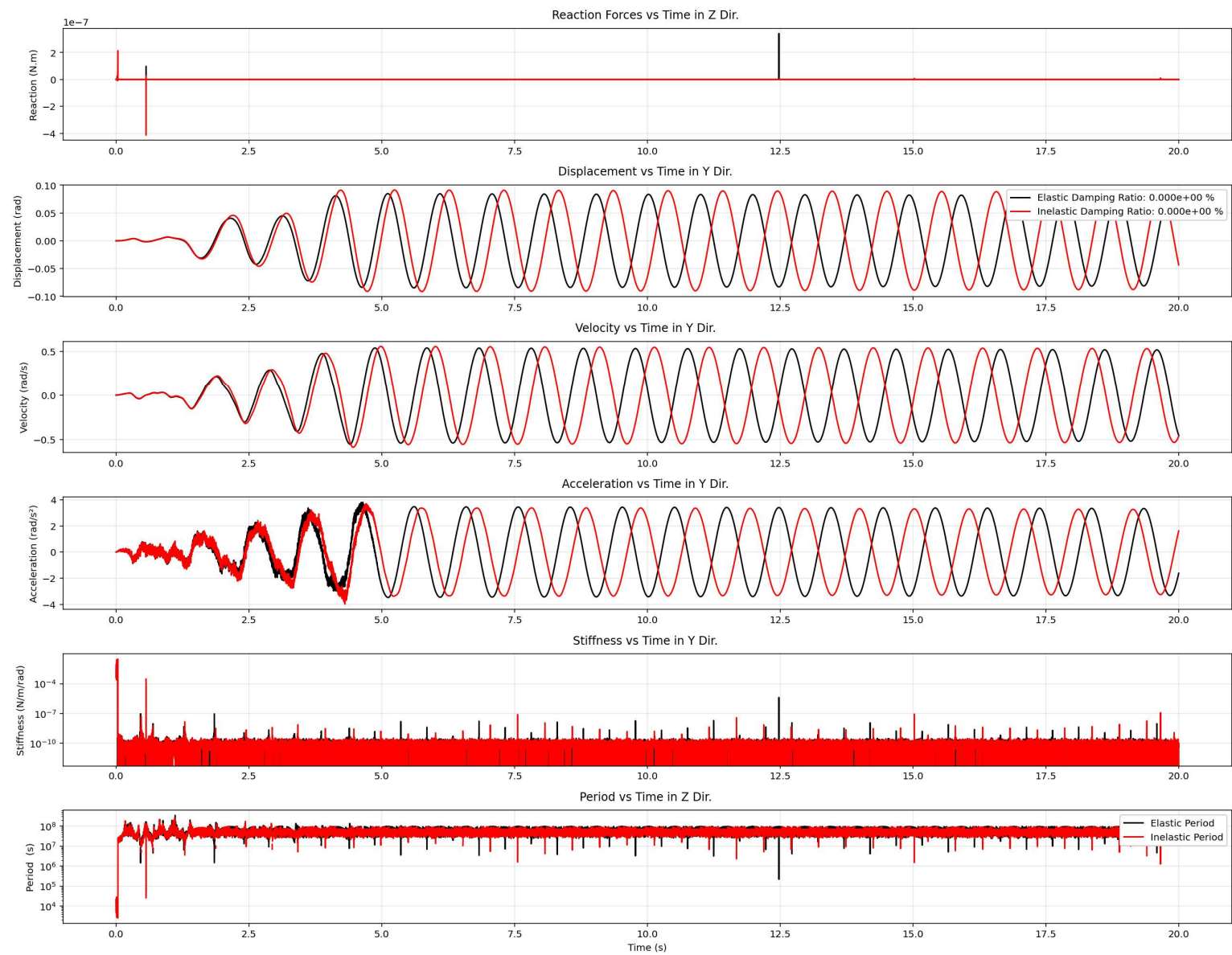


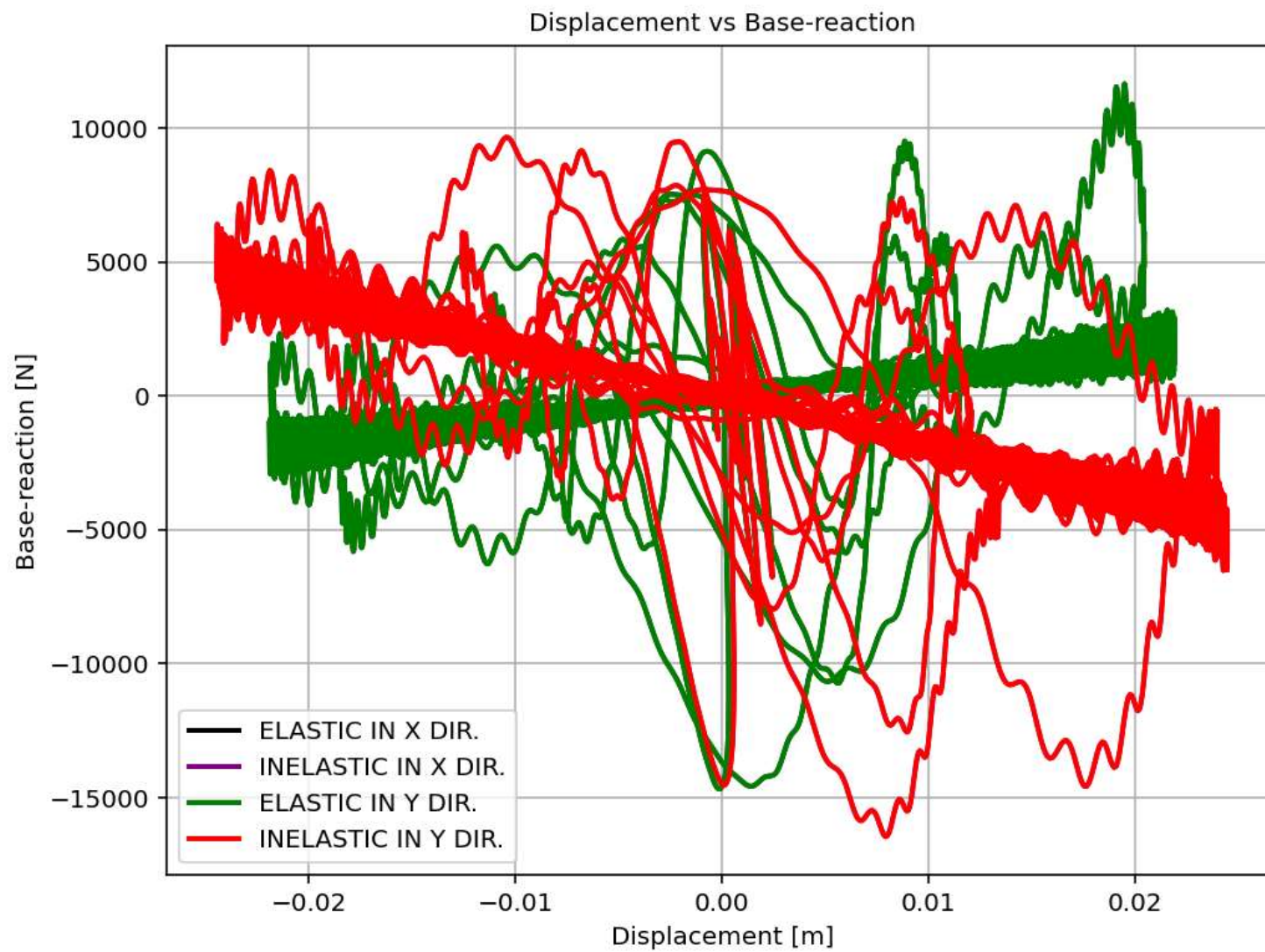


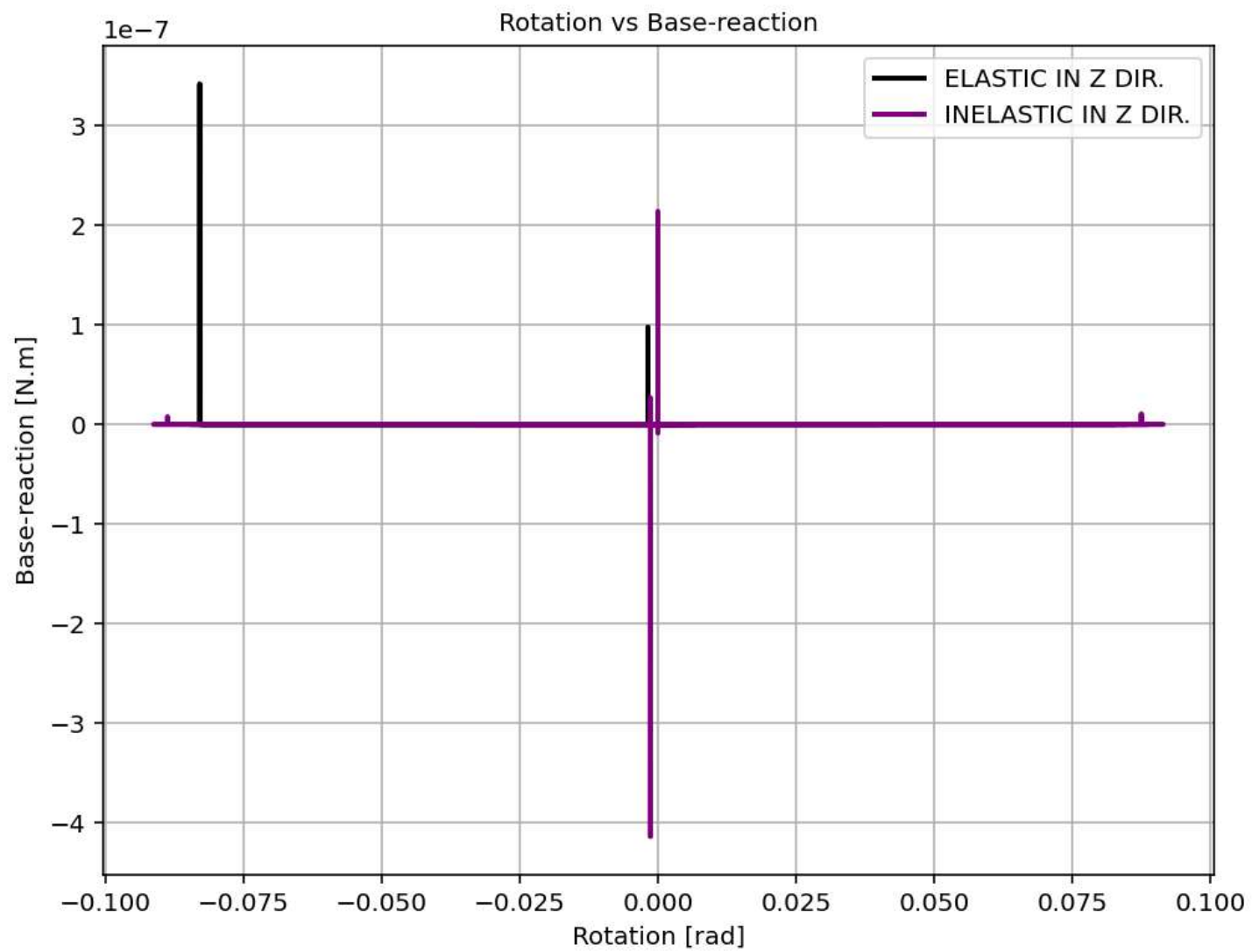


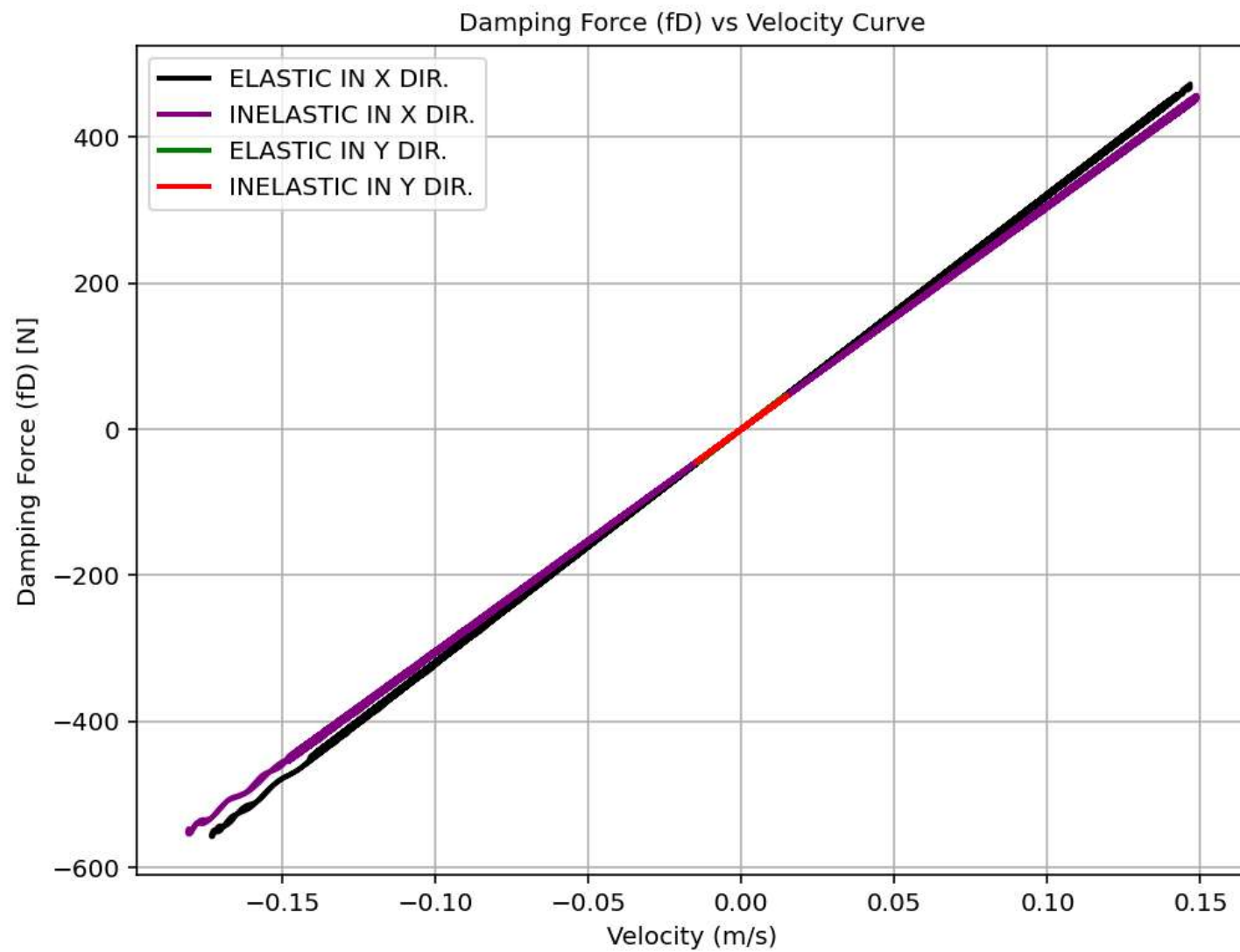


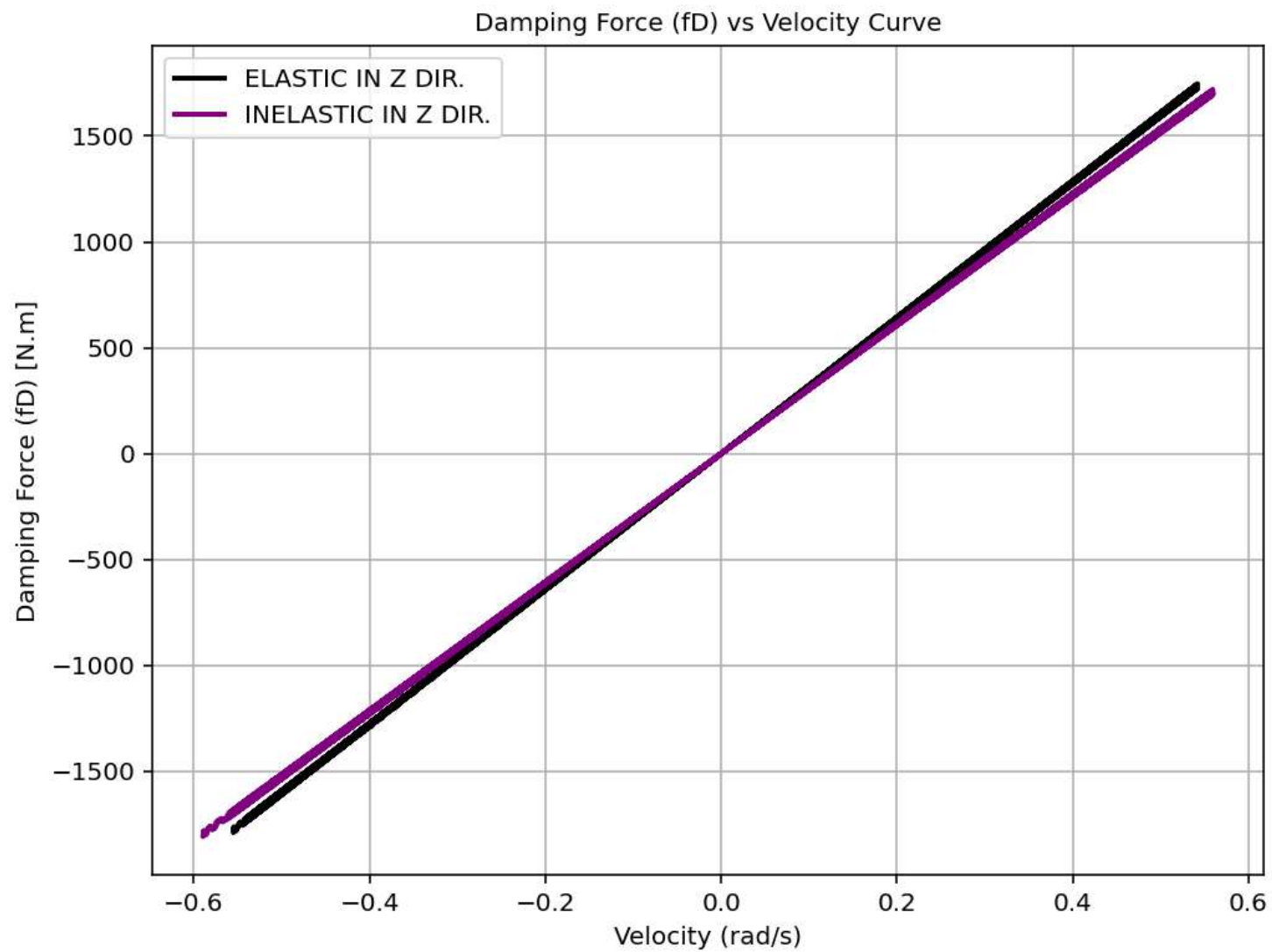


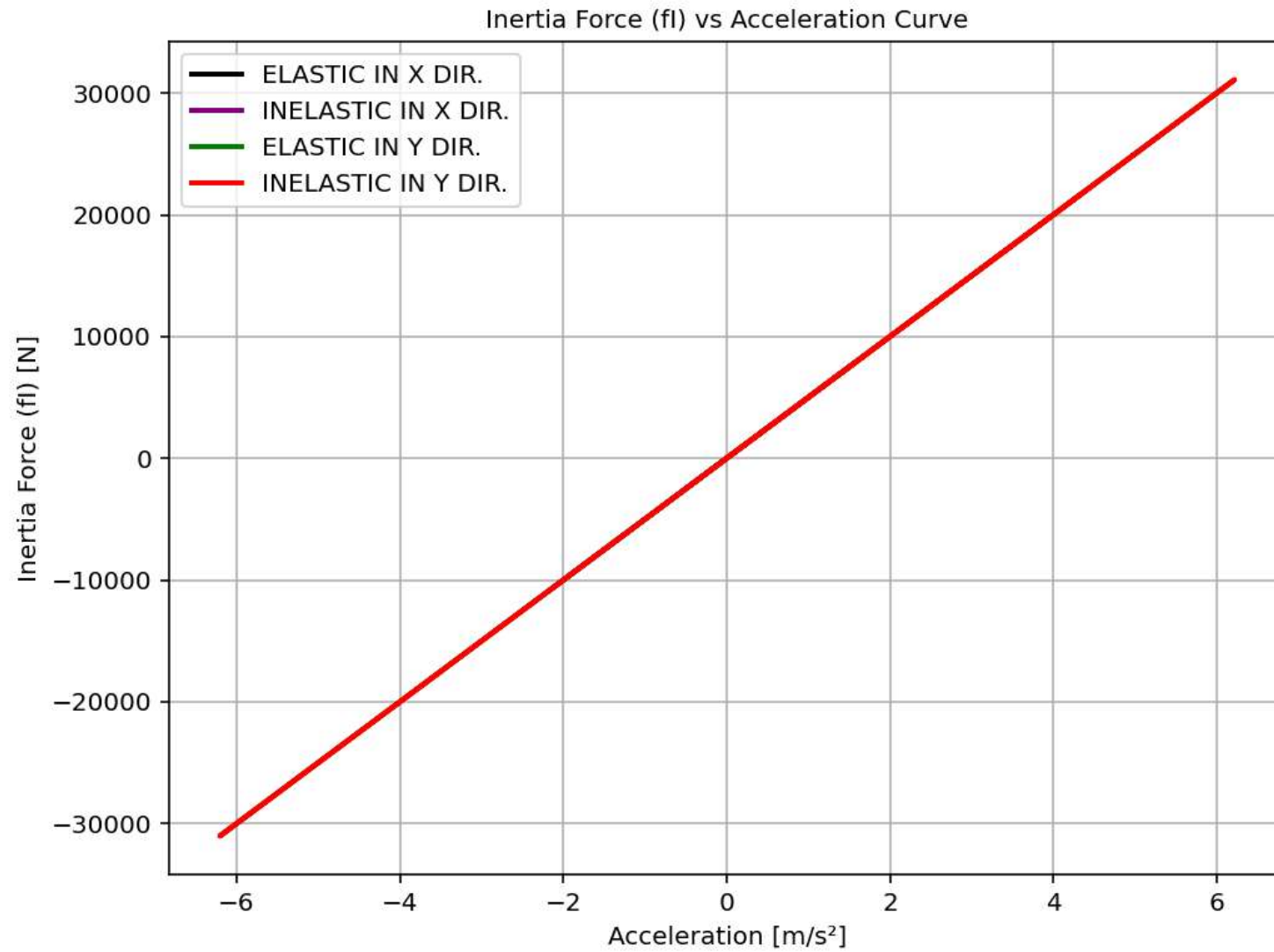


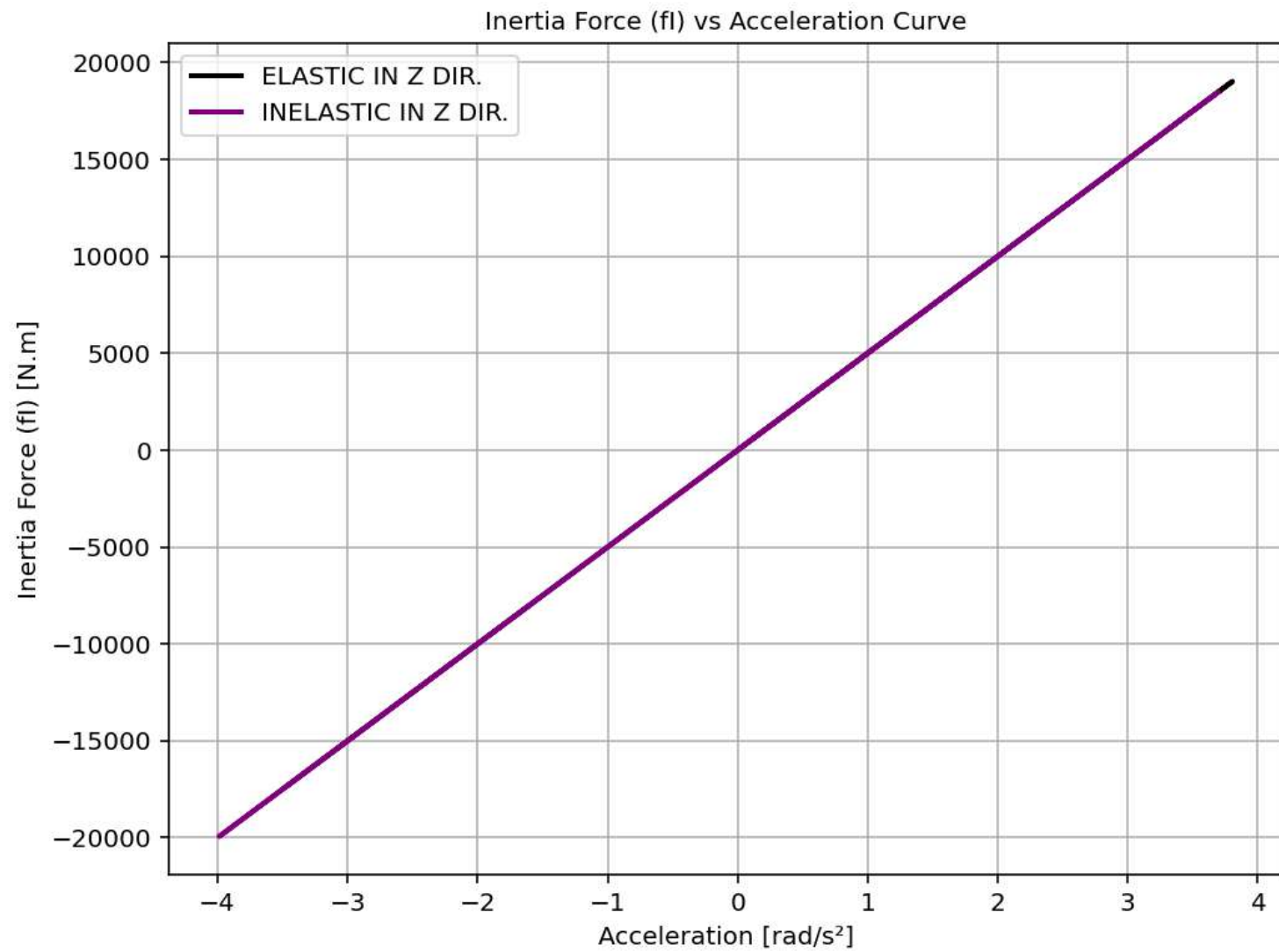


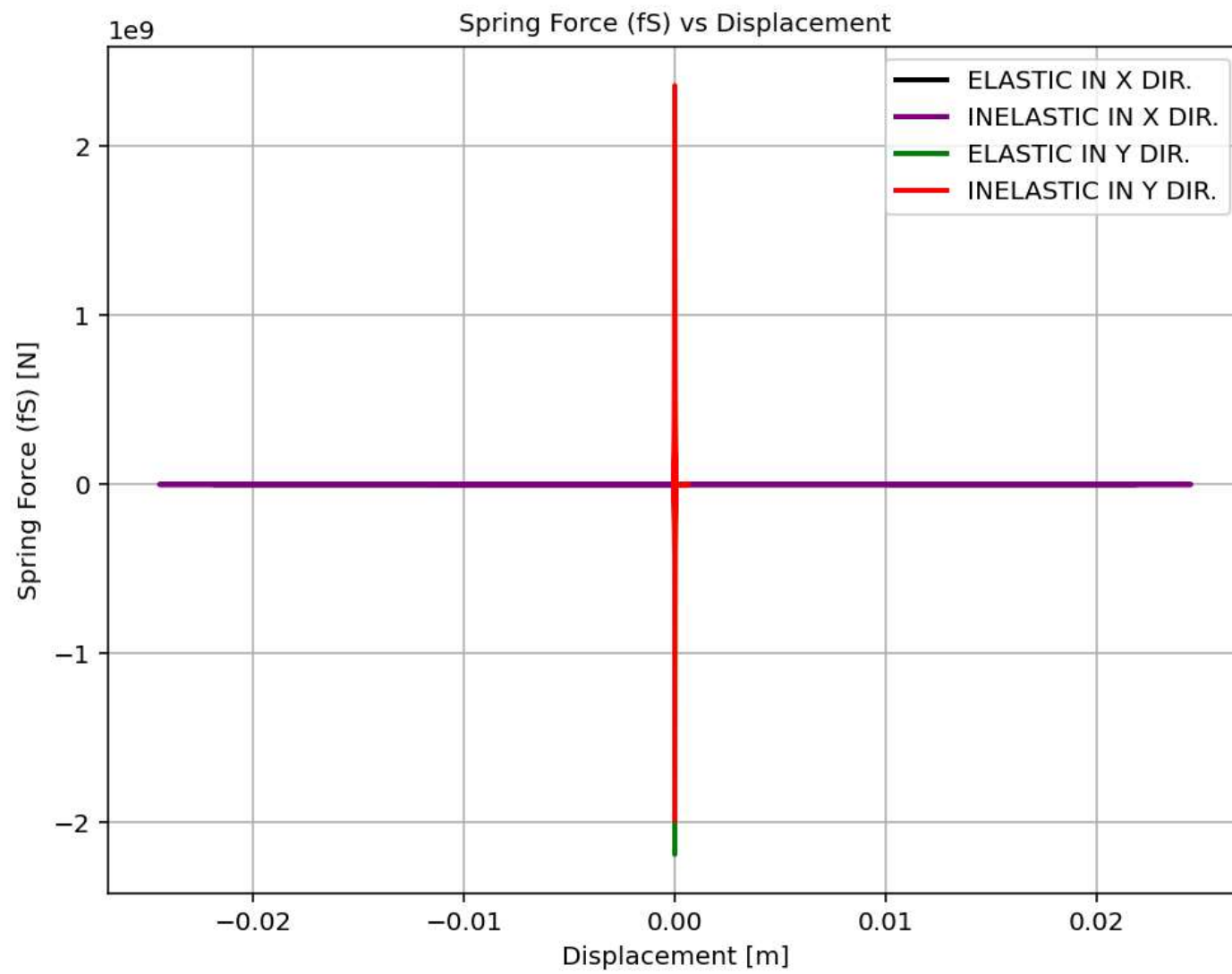




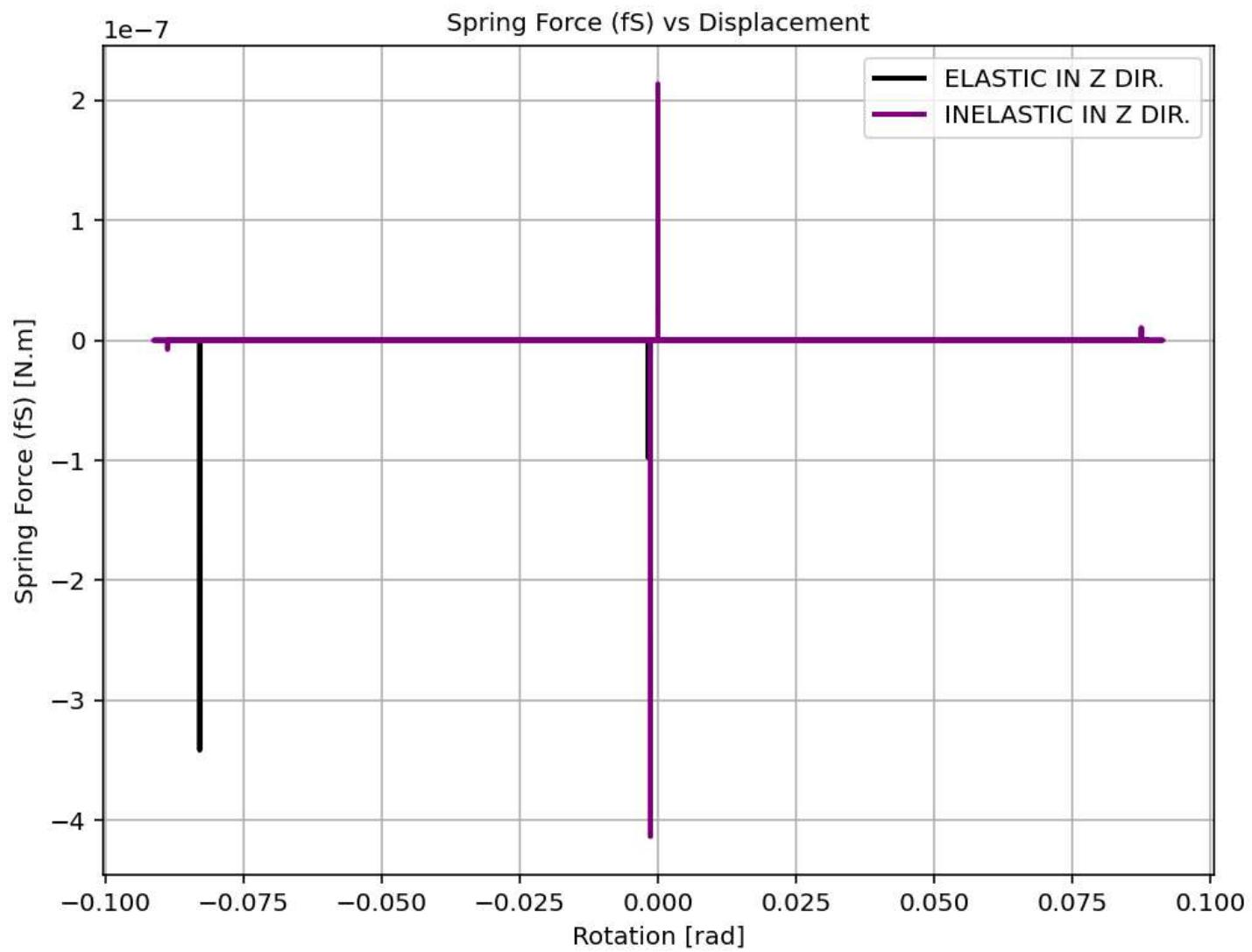


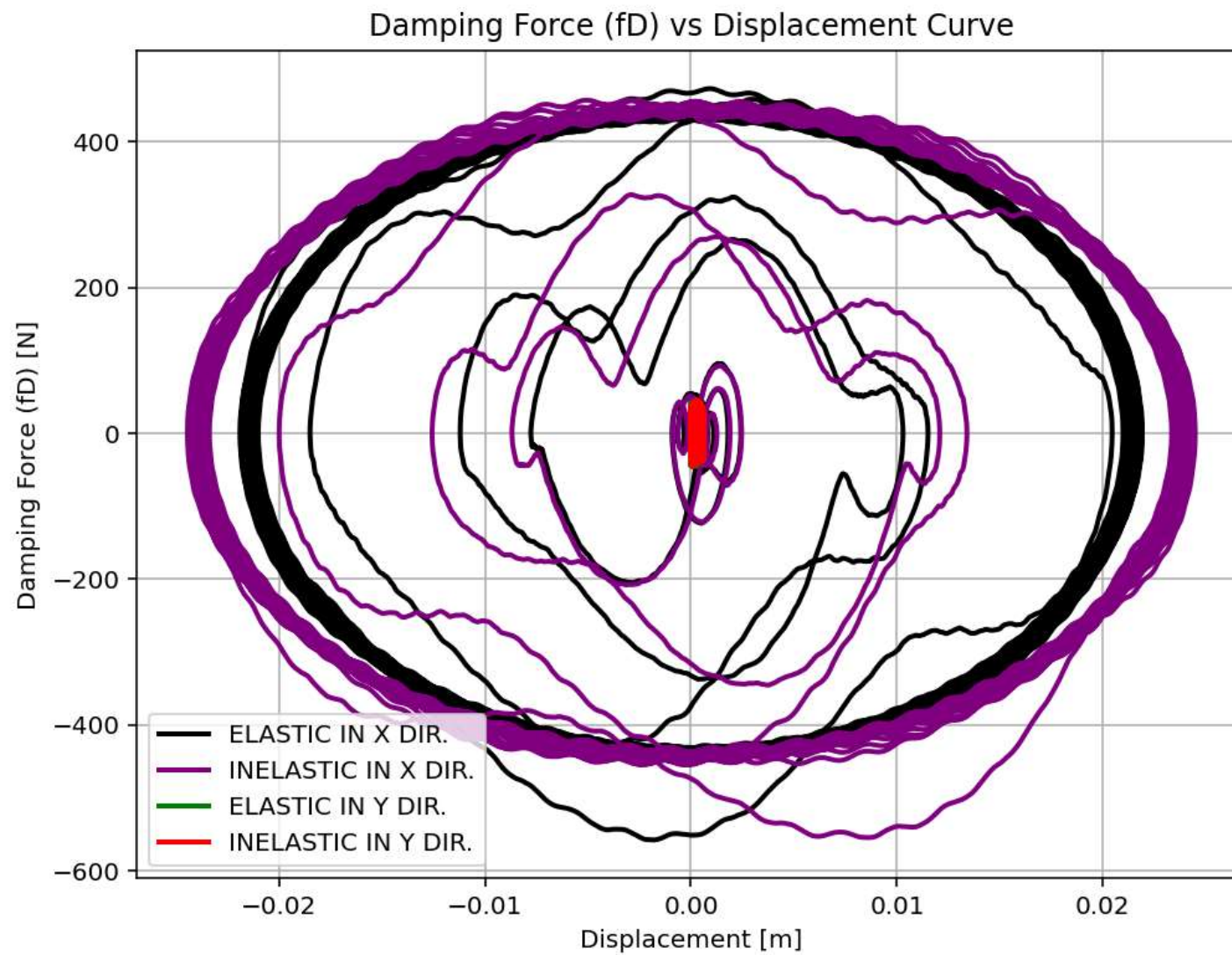












Damping Force ( $f_D$ ) vs Displacement Curve

